TTK4240 Industriell elektroteknikk

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Lecture - Week 04 Part II

- Review of the fundamentals in circuit analysis with sinusoidal signals
  - The concept of Phasor
- Current, voltage and power calculations
  - Instantaneous power
  - Average (real) power
  - Reactive power
  - Complex power
  - Power factor
  - Maximum real power delivered to a load
  - Load impedance required to deliver maximum real power to the load

Review and practical examples for Chapter 9 and 10

Phasor and property

Review

Definition

- Phasor is complex number defined at a specific frequency being used for representing a sinusoid.

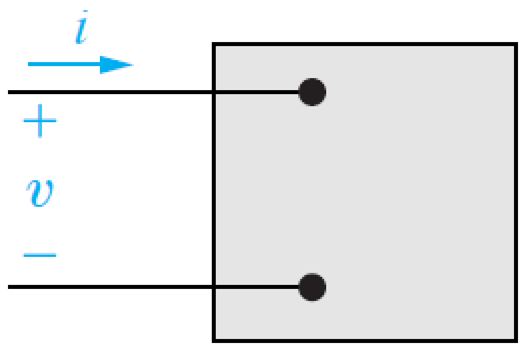
$$\underline{Ae^{j\theta}} \text{ at } \omega \quad \Longleftrightarrow \quad A\cos(\omega t+\theta)$$

Property

$$\underbrace{Ae^{j\theta}}_{\text{at } \omega} \Rightarrow \boxed{H(s)} \Rightarrow \underbrace{H(j\omega)\cdot Ae^{j\theta}}_{\text{at } \omega}$$

Review

Active and reactive power



$$v = V_m \cos(\omega t + \theta_v),$$

$$i = I_m \cos(\omega t + \theta_i),$$

Instantaneous power is defined as $p = vi.$

$$p = P + P\cos 2\omega t - Q\sin 2\omega t,$$

Active power

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i),$$

Reactive power

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i).$$

Apparent power

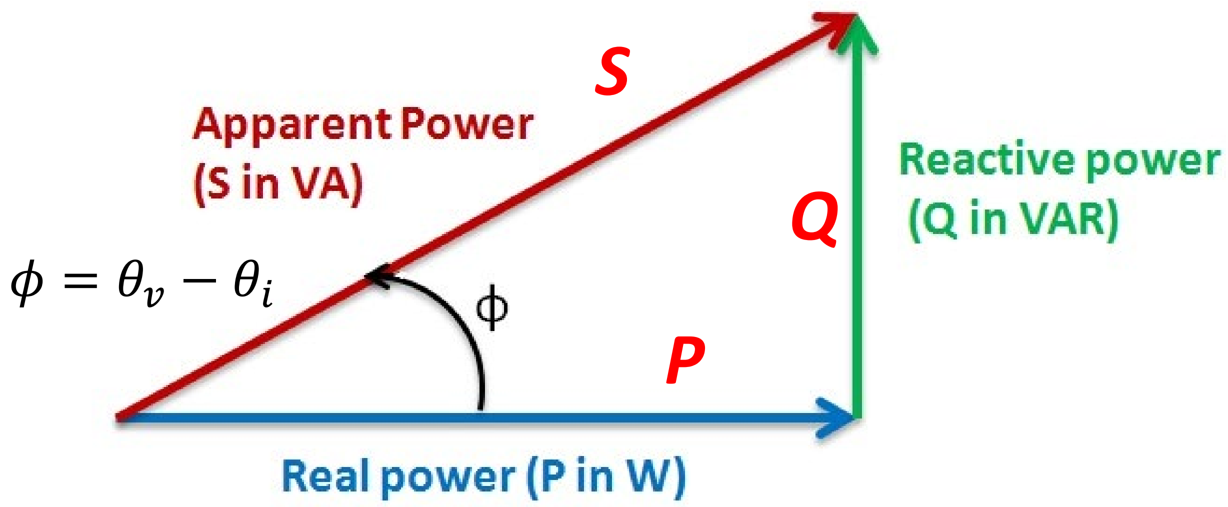
Review

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)$$

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)$$

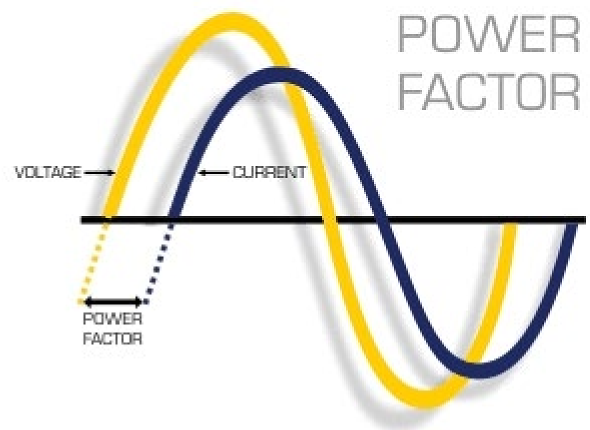
$$\sqrt{P^2 + Q^2} = \frac{V_m I_m}{2} = S$$

Apparent power

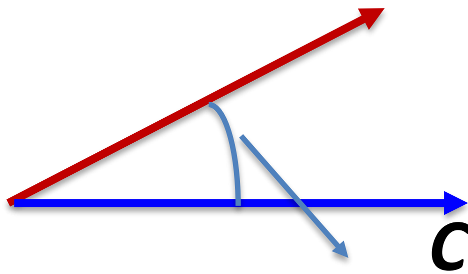


Power factor

Review



Voltage phasor U



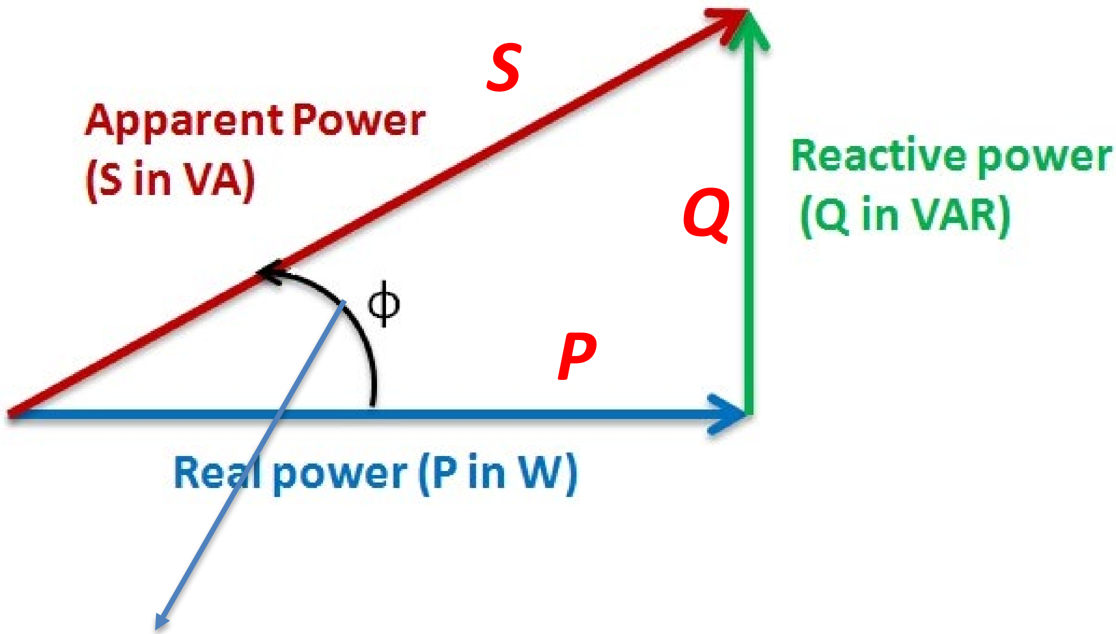
Current phasor I

Angle difference $\theta_v - \theta_i$

Power factor $\mathrm{pf} = \cos(\theta_v - \theta_i),$

Power factor

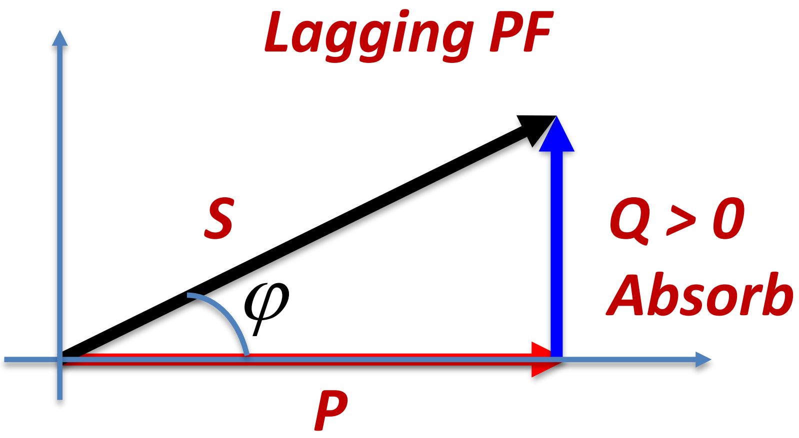
Review



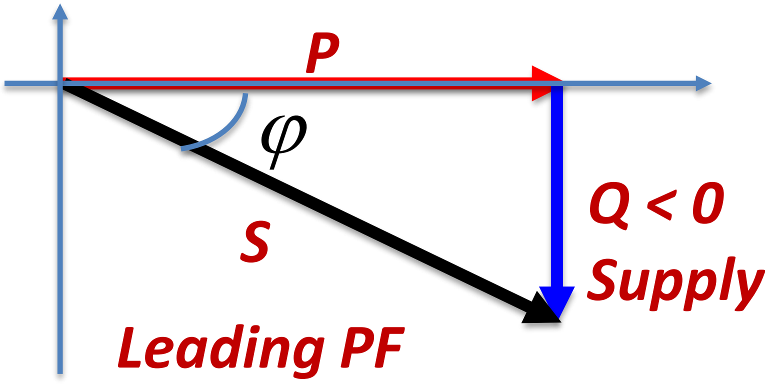
$$\cos\left(\theta_v - \theta_i\right) = \cos\varphi = \frac{P}{S}$$

Leading and lagging PF

The ***lagging PF*** refers to the current flowing through the circuit will be **lagging** the voltage (i.e. ***the Q > 0***)

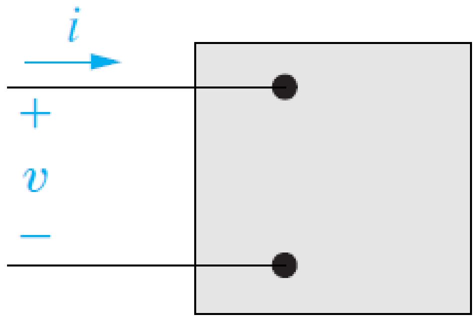


The ***leading PF*** refers to the current flowing through the circuit will be **leading** the voltage (i.e. ***the Q < 0***)



Practical Example of Application

Calculate Average and Reactive Power



$$v = 100\cos(\omega t + 15^\circ)\ \text{V},$$

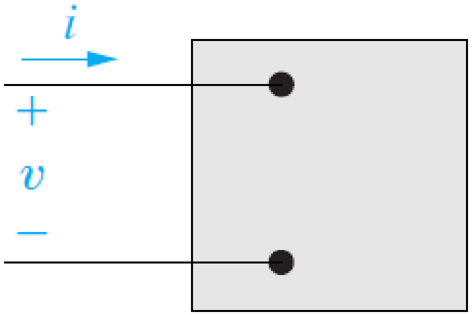
$$i = 4\sin(\omega t - 15^\circ)\ \text{A}.$$

a) Calculate the average power and the reactive power
b) Is the network absorbing or delivering power?
c) Is the network inside the box absorbing or supplying reactive power?

Question a) Calculate the average power and the reactive power ?

$$v = 100\cos\left(\omega t + 15^\circ\right)$$

$$\boxed{i = 4\sin\left(\omega t - 15^\circ\right)}$$



P and Q?

$$i = \underline{4}\cos\left(\omega t \underline{-15^\circ - 90^\circ}\right)$$

$$v = \underline{100}\cos\left(\omega t + \underline{15}^\circ\right)$$

$\theta_i = -105^\circ$ **Current phase**

$\theta_v = 15^\circ$ **Voltage phase**

$$V_m = 100$$

$$I_m = 4$$

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i),$$

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i).$$

Question a) Calculate the average power and the reactive power ?

$$\theta_i = -105^\circ$$

$$\theta_v = 15^\circ$$

$$V_m = 100$$

$$I_m = 4$$

Active power

$$P = \frac{4*100}{2}\cos\left(15^\circ - \left(-105^\circ\right)\right)$$

$$= 200\cos\left(120^\circ\right) = -100$$

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i),$$

Reactive power

$$Q = \frac{4*100}{2}\sin\left(15^\circ - \left(-105^\circ\right)\right)$$

$$= 200\sin\left(120^\circ\right) = 100\sqrt{3}$$

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i).$$

A check

$$\sqrt{P^2 + Q^2} = S$$

$$S = \frac{100 \cdot 4}{2} = 200$$

$$\sqrt{P^2 + Q^2} = \sqrt{\left(-100\right)^2 + \left(100\sqrt{3}\right)^2} = 200$$

$$\theta_i = -105^\circ$$

$$\theta_v = 15^\circ$$

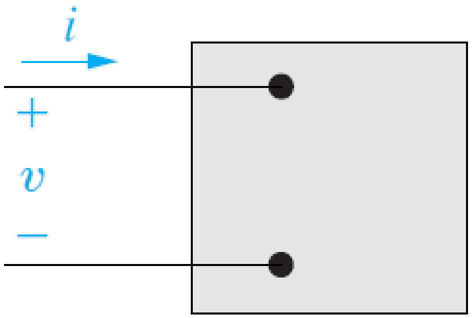
$$V_m = 100$$

$$I_m = 4$$

Question b) and c)

b) Is the network absorbing or delivering power?

Delivering



c) Is the network inside the box absorbing or supplying reactive power (magnetizing vars)?

$$P = -100$$

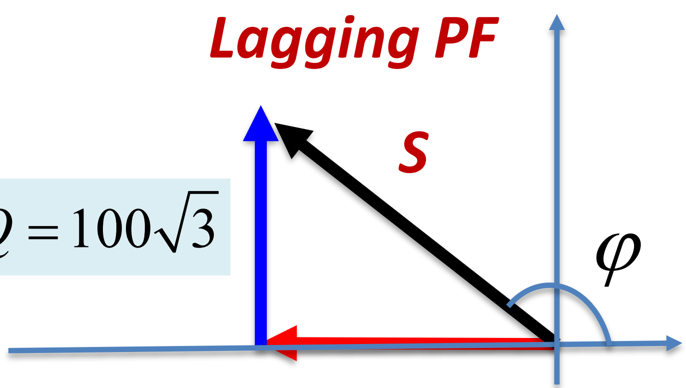
$$Q = 100\sqrt{3}$$

Absorbing

What is the PF?

$$\theta_i = -105^\circ, \theta_v = 15^\circ$$

$$\cos(\theta_v - \theta_i) \Rightarrow = \cos\left(15 - \left(-105^\circ\right)\right) = \cos\left(120^\circ\right) = -\frac{1}{2}$$



$$Q = 100\sqrt{3}$$

$$P = -100$$

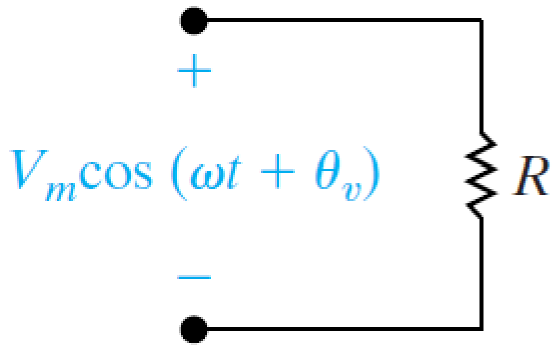
$$\cos(\varphi) = \frac{P}{S} = \frac{-100}{200} = \boxed{-}\frac{1}{2}$$

The negative sign indicates the active power is inversed (delivery)

Review of the RMS value calculation

RMS Value and Power calculations

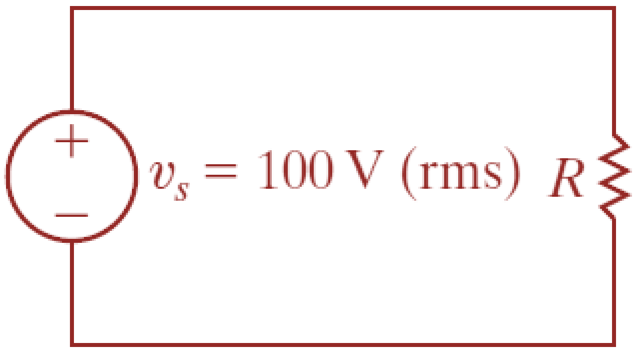
AC system



$$P = \frac{1}{T}\int_{t_0}^{t_0+T} \frac{V_m^2 \cos^2(\omega t + \phi_v)}{R} dt$$

$$= \frac{1}{R}\underbrace{\left[\frac{1}{T}\int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi_v) dt\right]}_{V_{rms}^2}.$$

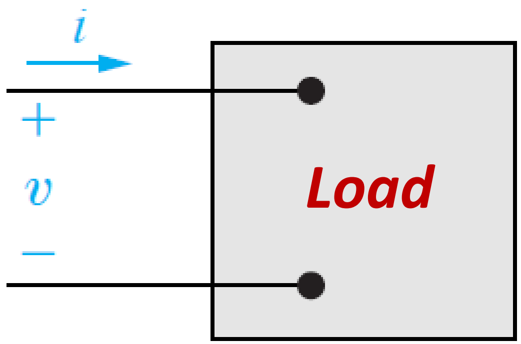
DC system



Equivalent

$$P = \frac{V_{\text{rms}}^2}{R}.$$

RMS Value and Power calculations



$$v = V_m \cos(\omega t + \theta_v)$$

$$i = I_m \cos(\omega t + \theta_i)$$

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i),$$

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

$\theta_v = \theta_i$ Is the case with a pure resistive load

$\theta_v \neq \theta_i$ Some capacitors or inductors exist in the system

Effective Value of Power

Effective value of active power

$$P = \frac{V_m I_m}{2}\cos\left(\theta_v - \theta_i\right)$$

$$= \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos\left(\theta_v - \theta_i\right)$$

$$= V_{eff} I_{eff} \cos\left(\theta_v - \theta_i\right)$$

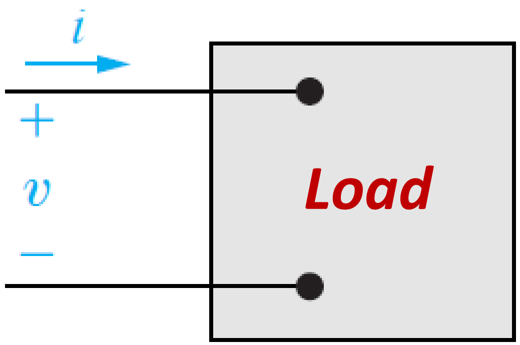
Effective value of reactive power

$$Q = \frac{V_m I_m}{2}\sin\left(\theta_v - \theta_i\right)$$

$$= \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\sin\left(\theta_v - \theta_i\right)$$

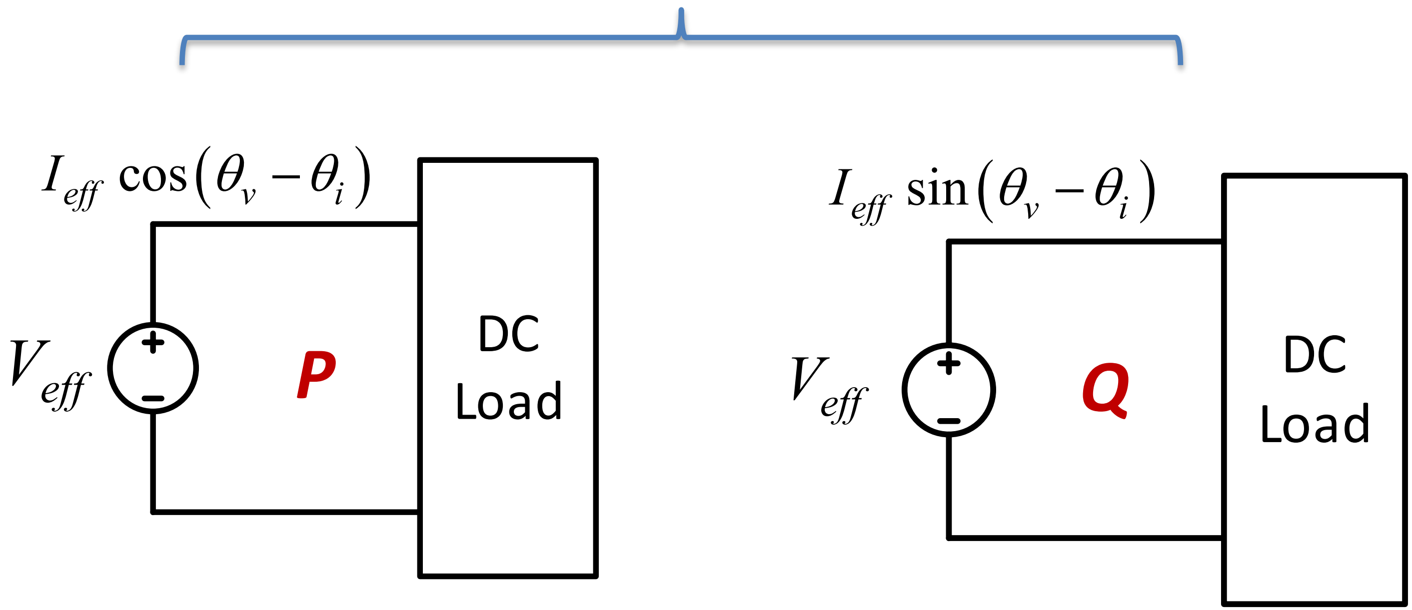
$$= V_{eff} I_{eff} \sin\left(\theta_v - \theta_i\right)$$

Equivalent DC systems



$$v = V_m \cos\left(\omega t + \theta_v\right)$$

$$i = I_m \cos\left(\omega t + \theta_i\right)$$

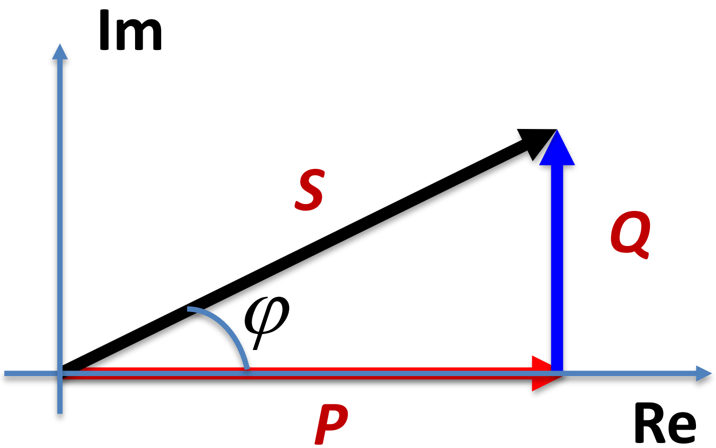


Reactive Power

$$Q = V_{\text{eff}} I_{\text{eff}} \sin(\theta_v - \theta_i).$$

- The rating of equipments depends on Q
- The magnetic and electric fields of inductors and capacitors store energy related to Q

Complex Power



Complex power

$$\boxed{S} = P + jQ$$

$$P = V_{eff} I_{eff} \cos\left(\theta_v - \theta_i\right)$$

$$Q = V_{eff} I_{eff} \sin\left(\theta_v - \theta_i\right)$$

$$S = V_{eff} I_{eff} \left(\cos\left(\theta_v - \theta_i\right) + j\sin\left(\theta_v - \theta_i\right)\right)$$

Complex power and phasors

$$S = V_{eff} I_{eff} \underline{\left(\cos\left(\theta_v - \theta_i\right) + j\sin\left(\theta_v - \theta_i\right)\right)}$$

Applying Euler identity

$$S = V_{eff} I_{eff} \underline{e^{j\left(\theta_v - \theta_i\right)}}$$

$$= \underbrace{\left[V_{eff} e^{j\theta_v}\right]}_{\text{Voltage phasor } V} \cdot \underbrace{\left[I_{eff} e^{-j\theta_i}\right]}_{\text{Conjugate of current phasor } I}$$

Voltage phasor *V*

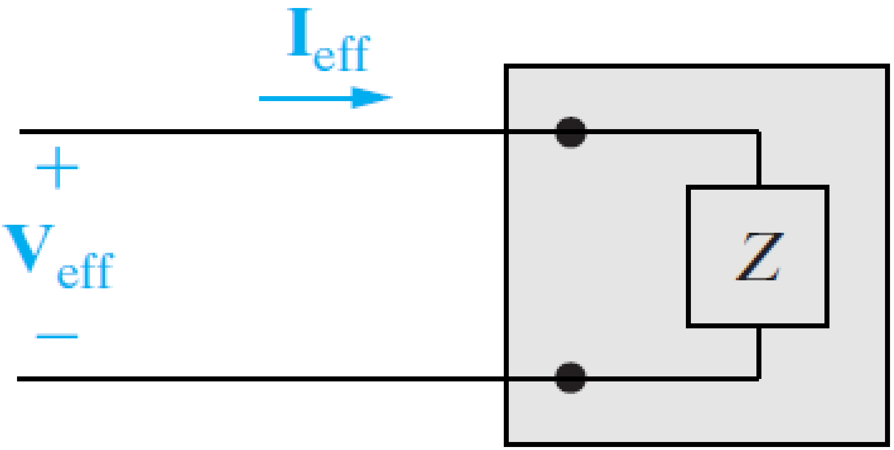
Conjugate of current phasor *I*

Complex power represented by voltage and current phasors

$$S = V_{eff} \cdot I^*_{eff}$$

$$S = \frac{1}{2} V_m \cdot I^*_m$$

Complex power with Impedances



$$S = V_{eff} I_{eff}^*$$

$$V_{eff} = Z I_{eff}$$

Replace current with voltage

$$S = V_{eff}\left(\frac{V_{eff}}{Z}\right)^* \Rightarrow S = \frac{\left|V_{eff}\right|^2}{Z^*}$$

Replace voltage with current

$$S = Z I_{eff} I_{eff}^* \Rightarrow S = Z\left|I_{eff}\right|^2$$

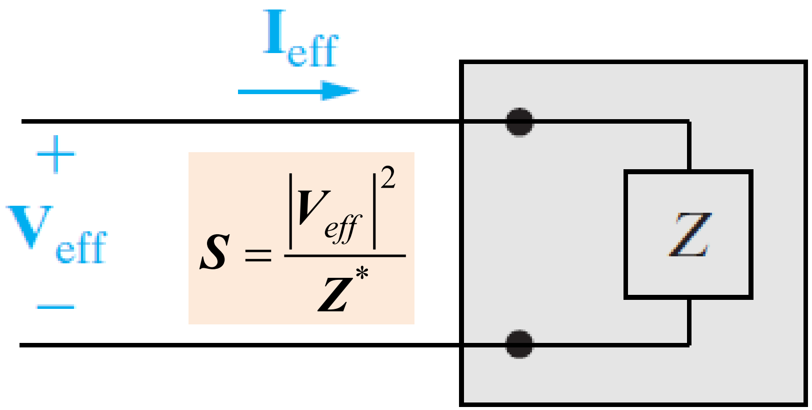
Complex power of R, L, C

Resistor

$$Z = R$$

$$S = \frac{|V_{eff}|^2}{R} = P$$

P>0



Inductor

$$Z = jX\left(X > 0\right)$$

$$S = \frac{|V_{eff}|^2}{-jX} = j\frac{|V_{eff}|^2}{X} = jQ$$

Q>0

Capacitor

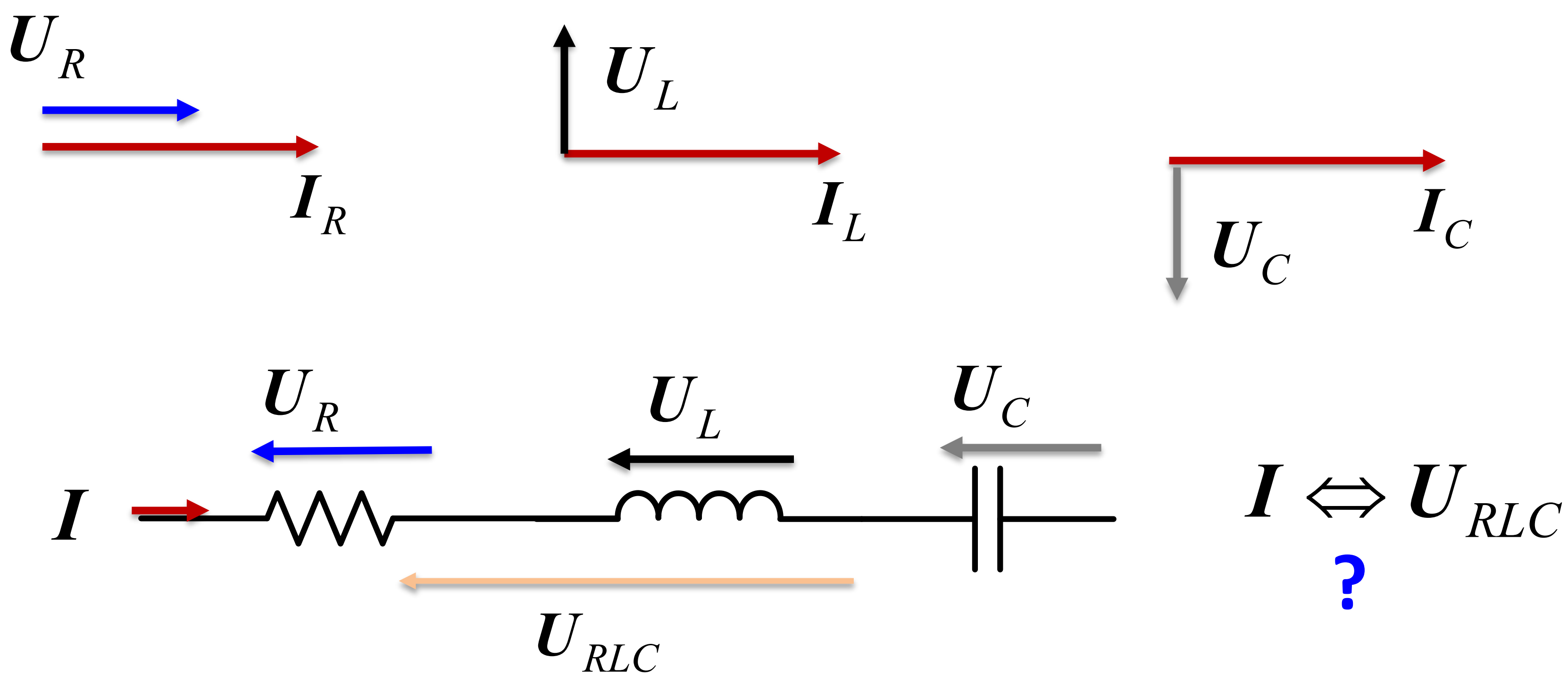
$$Z = jX\left(X < 0\right)$$

$$S = \frac{|V_{eff}|^2}{-jX} = j\frac{|V_{eff}|^2}{X} = jQ$$

Q<0

Phasor diagram

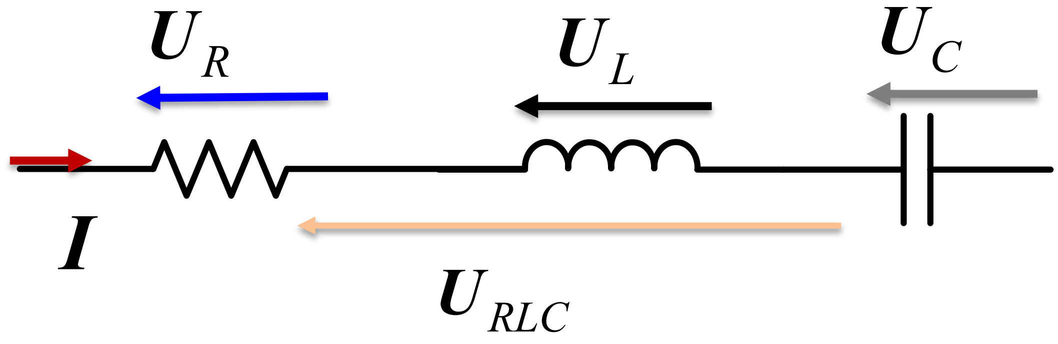
Phasor Diagram is a graphical way of representing the magnitude and directional relationship between two or more phasors.

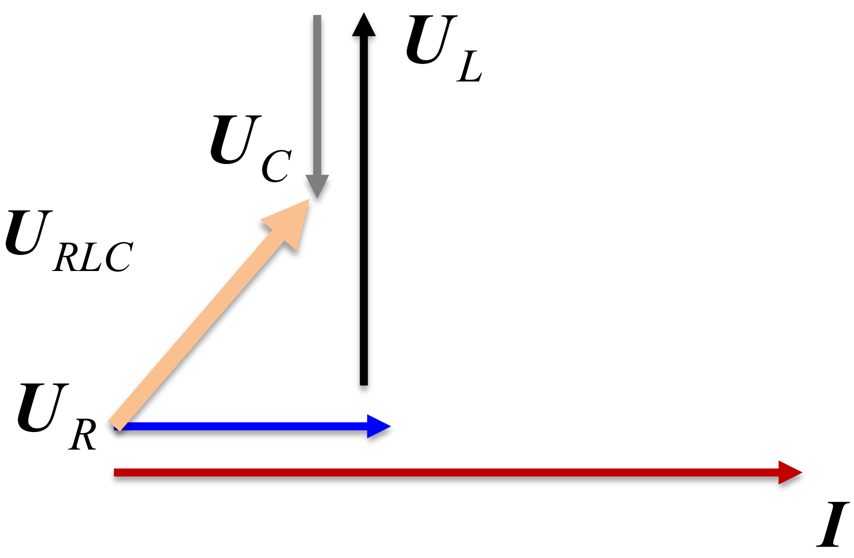


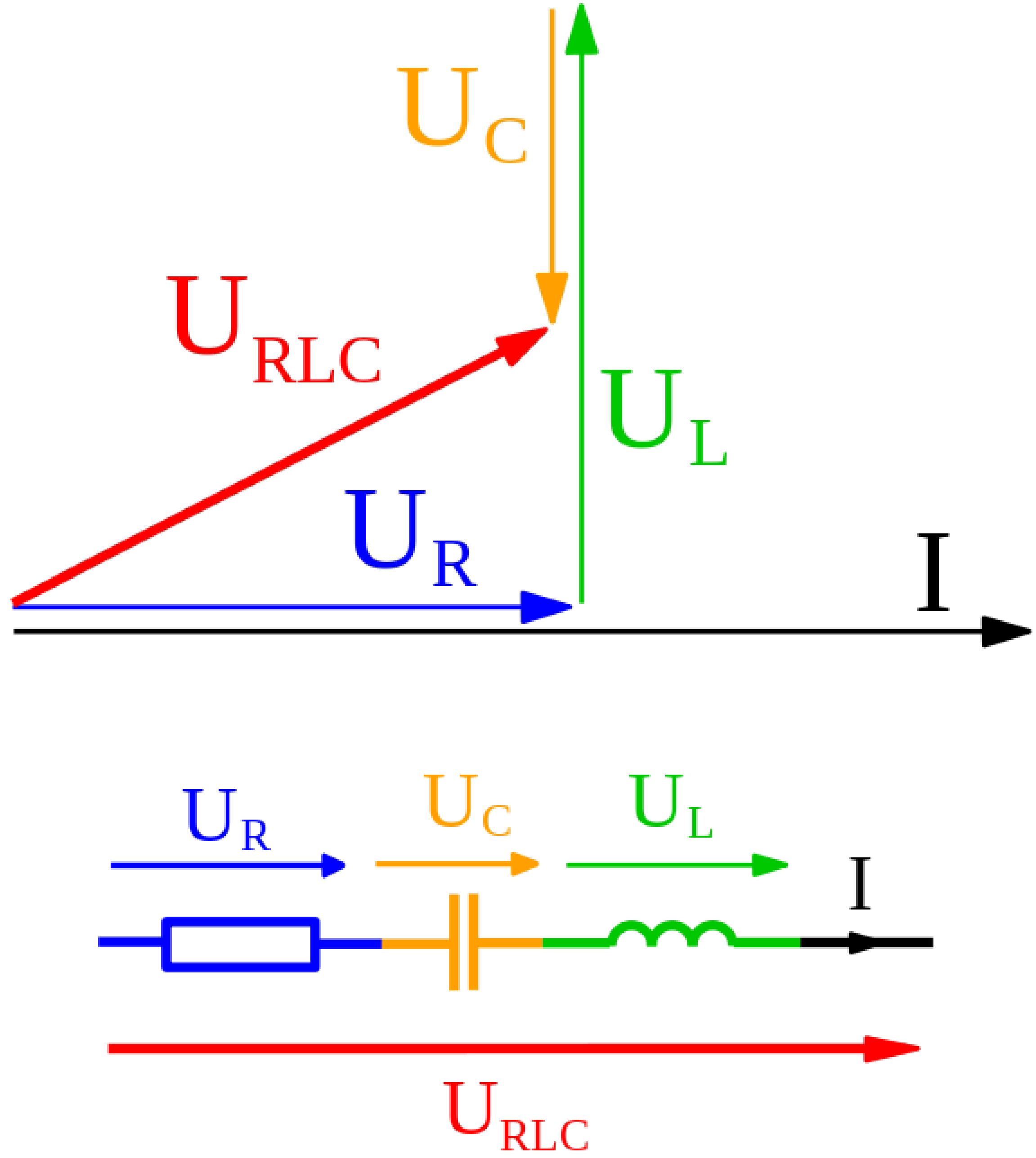
Phasor diagram of a series connected RLC circuit

$$U_{RLC} = U_R + U_L + U_C$$

$$I = I_R = I_L = I_C$$






U_C
U_{RLC}
U_L
U_R
I
U_R
U_C
U_L
I
U_{RLC}

Practical Example of Application

Appliances consumption

Appliance	Average Wattage	Est. kWh Consumed Annually[a]
Food preparation		
Coffeemaker	1200	140
Dishwasher	1201	165
Egg cooker	516	14
Frying pan	1196	100
Mixer	127	2
Oven, microwave (only)	1450	190
Range, with oven	12,200	596
Toaster	1146	39

$$S = V_{eff} I_{eff}^{*}$$

$$P = V_{eff} I_{eff}$$

If the kitchen outlets (120 V) are protected by a 20A fuse, and assuming that the Coffee maker, egg cooker, frying pan and toaster are in operation at the same time:

Will the circuit be interrupted by the protecting device?

Calculation

The total power of Coffee maker, egg cooker, frying pan and toaster is

P =
1200 +516+1196+1146
= 4058 W

Appliance	Average Wattage	Est. kWh Consumed Annually[a]
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$$\left|I_{eff}\right| = \left|\frac{S^*}{V_{eff}^*}\right| = \frac{P}{V_{eff}} = \frac{4058\ W}{120\ V} = 33.8A > 20A$$

The circuit will be interrupted

Practical Example

Calculate Complex Power

An electrical load operates at 240 V rms, absorbing 8 kW at a lagging power factor of 0.8:

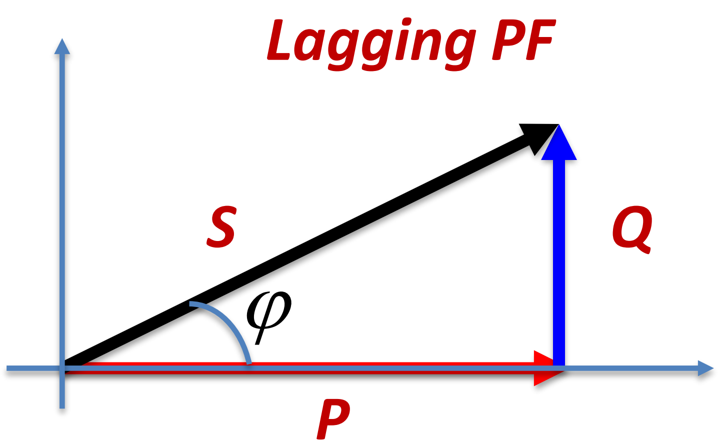
a) Calculate the complex power of the load
b) Calculate the impedance of the load

$$U_{eff} = 240V$$

$$P = 8kW$$

$$\cos\varphi = 0.8$$

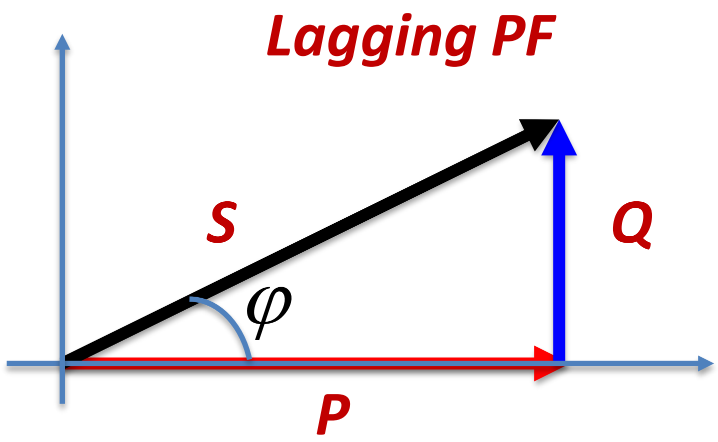
$$\varphi > 0$$



Calculate Complex Power

$$S = \frac{P}{\cos\varphi} = \frac{8kW}{0.8} = 10kW$$

$$Q = \pm\sqrt{S^2 - P^2}$$
$$= \pm\sqrt{10^2 - 8^2} = \pm 6kVars$$



Lagging PF $\quad Q = 6kVars$

$$\boldsymbol{S} = P + jQ = (8 + 6j)\,kVA$$

Calculate the impedance

Recall the complex power $\boldsymbol{S} = \dfrac{\left|\boldsymbol{V}_{eff}\right|^2}{\boldsymbol{Z}^*}$

$$\boldsymbol{Z} = \frac{\left|\boldsymbol{V}_{eff}\right|^2}{\boldsymbol{S}^*} = \frac{\left|240V\right|^2}{\left(8-j6\right)kVA} = \frac{57.6}{8-j6}$$

$$= \frac{57.6}{10\angle -36.87^\circ} = 5.76\angle 36.87^\circ = 4.608 + j3.456$$

Exercise 1: use of RMS and Average P

- Problem 10.14:

a) Find the rms value of the periodic voltage shown in figure P10.14:

Figure P10.14

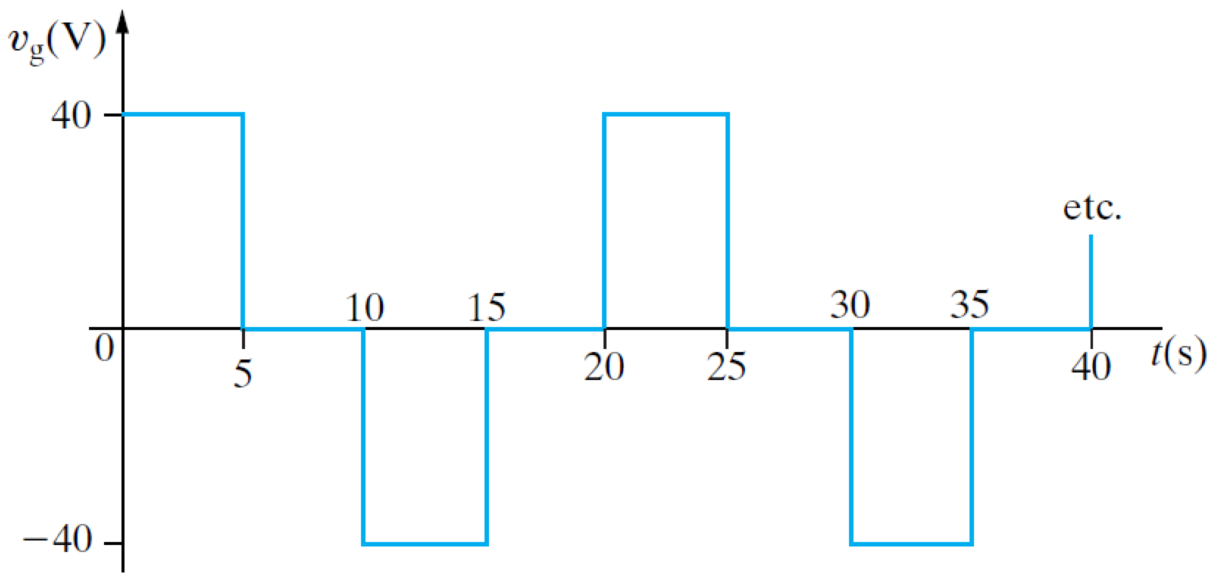
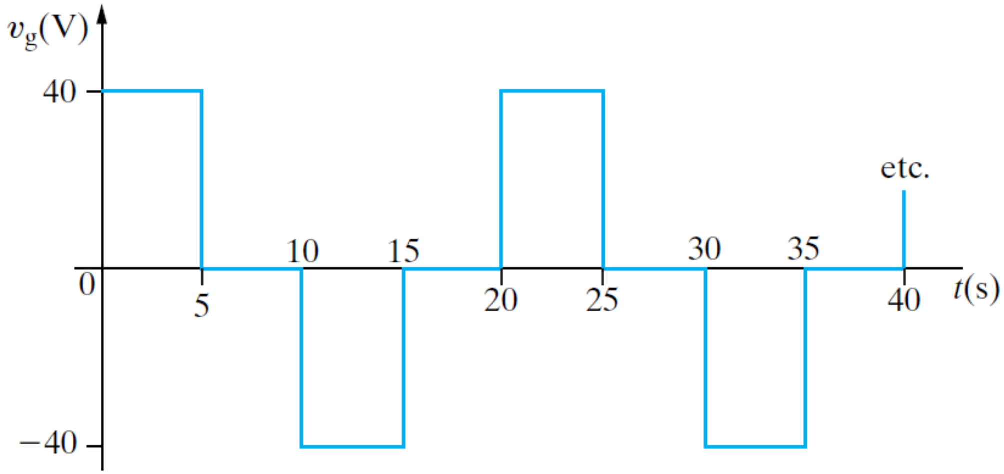


Figure P10.14



a) Find the rms value of the periodic voltage shown in figure.

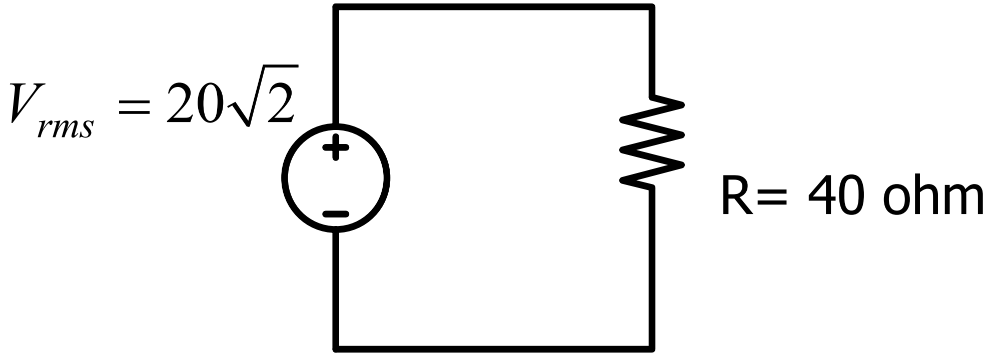
$$T = 20s$$

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v_g^2(t)\ dt} = \sqrt{\frac{1}{20}\left(\int_0^5 40^2\,dt + \int_{10}^{15}\left(-40\ \right)^2 dt\right)}$$

$$= \sqrt{\frac{1}{20}\left(5\cdot 40^2 + 5\cdot(-40)^2\right)}$$

$$= 20\sqrt{2}$$

b) Suppose the voltage in part a) is applied to the terminals of a 40 Ohm resistor. Calculate the average power dissipated by the resistor.



$$P = \frac{V_{rms}^2}{R} = \frac{\left(20\sqrt{2}\right)^2}{40} = 20W$$

c) When the voltage in part a) is applied to a different resistor, that resistor dissipates 10 mW of average power. What is the value of the resistor?

$$R = \frac{V_{rms}^2}{P} = \frac{\left(20\sqrt{2}\right)^2}{10\times10^{-3}} = 8\times10^4\, ohm$$

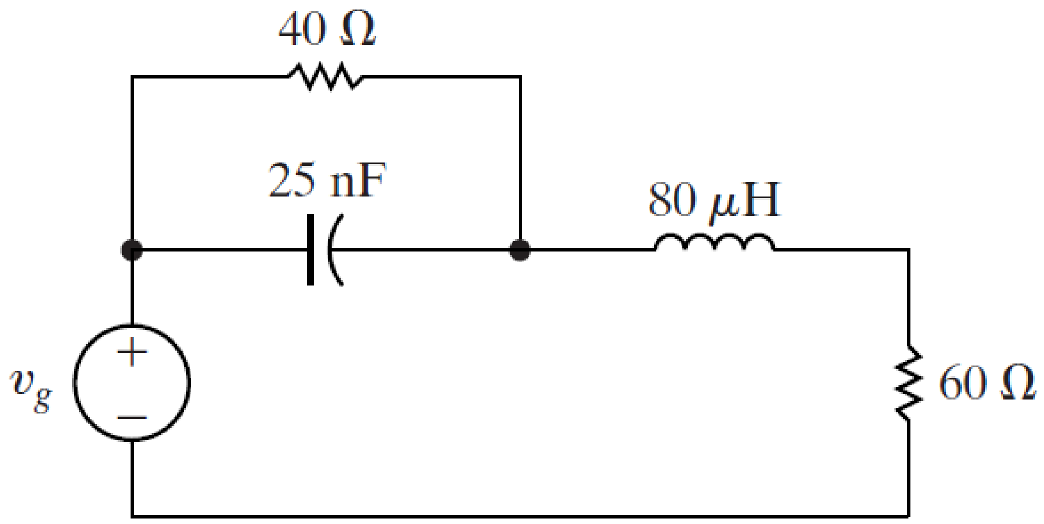
Exercise 2: use of Phasor

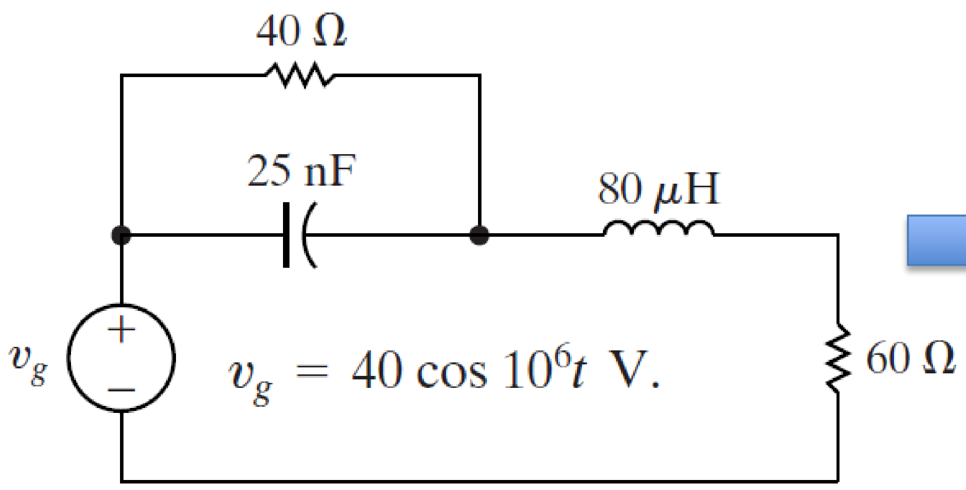
- Problem 10.20

a) Find the average power, the reactive power and the apparent power supplied by the voltage source in the circuit in figure P10.20.

Figure P10.20

$$v_g = 40 \cos 10^6 t \text{ V.}$$





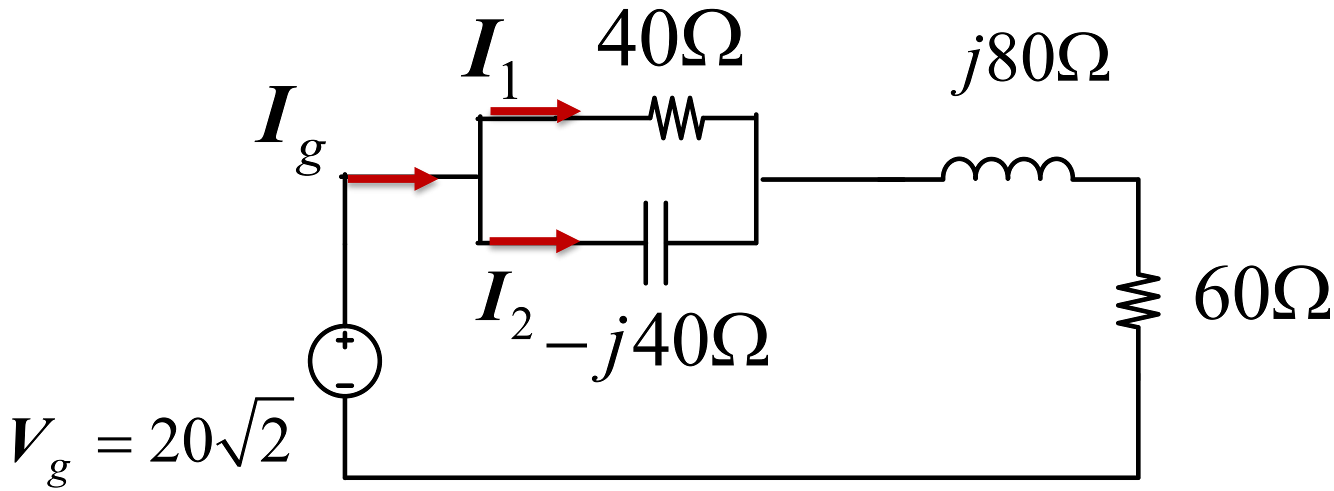
Step1:
Find out the frequency domain circuit

$$\omega = 10^6$$

$$-j\frac{1}{\omega C} = \frac{-j}{25nF \times 10^6} = -j40\Omega$$

$$j\omega L = j80uH \times 10^6 = j80\Omega$$

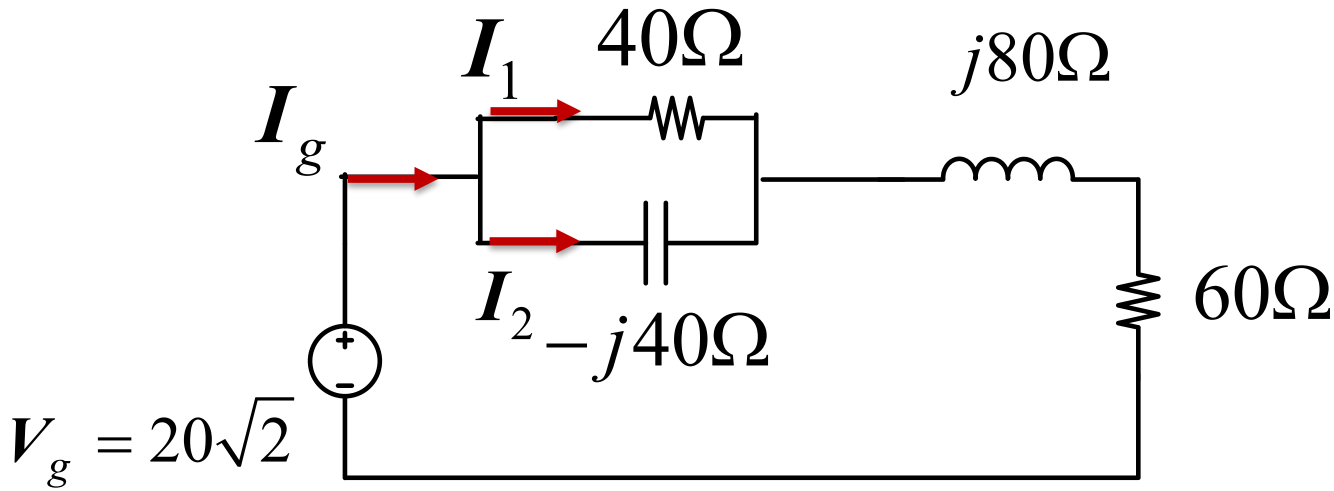
$$V_g = 20\sqrt{2}$$



a) Find the average power, the reactive power and the apparent power

Step2: Find the equivalent impedance

$$Z_{eq} = -j40\Omega \,||\, 40\Omega + j80\Omega + 60\Omega$$
$$= (80 + 60j)\Omega$$



a) Find the average power, the reactive power and the apparent power

Solving the current phasor

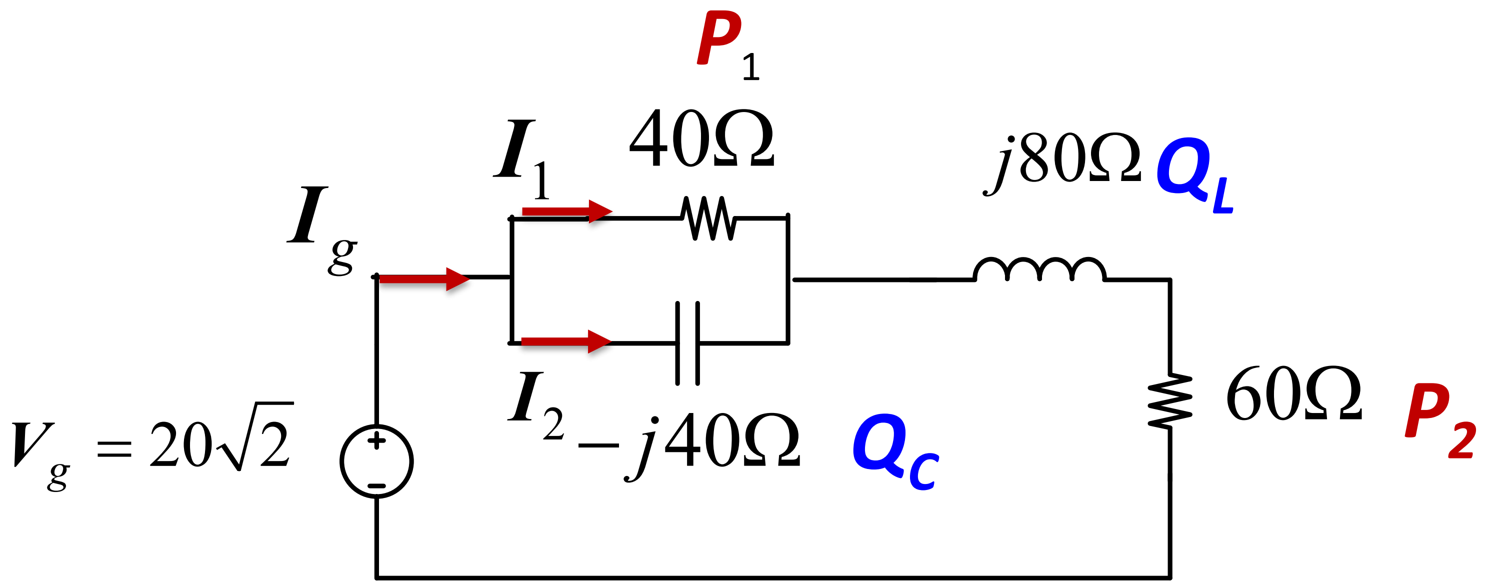
$$I_g = \frac{V_g}{Z_{eq}} = \frac{20\sqrt{2}}{(80+60j)} = (0.226 - j0.169)\,A$$

Solving the complex power

$$S = V_g I_g^* = \frac{|V_g|^2}{Z_{eq}^*} = \frac{\left(20\sqrt{2}\right)^2}{80-60j} = 8\angle 36.8^\circ = (\underbrace{6.4}_{P_{dev}} + j\underbrace{4.8}_{Q_{abs}})VA$$

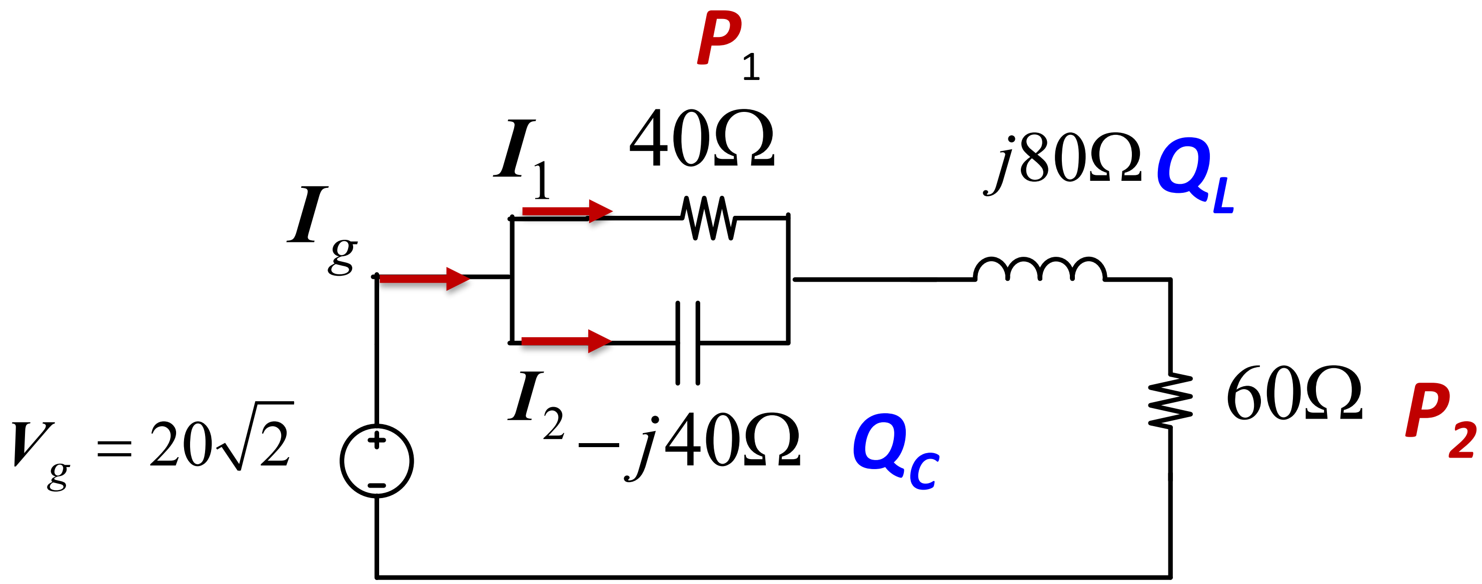
b) Check your answer in (a) by showing $P_{\text{dev}} = \sum P_{\text{abs}}$.

c) Check your answer in (a) by showing $Q_{\text{dev}} = \sum Q_{\text{abs}}$.



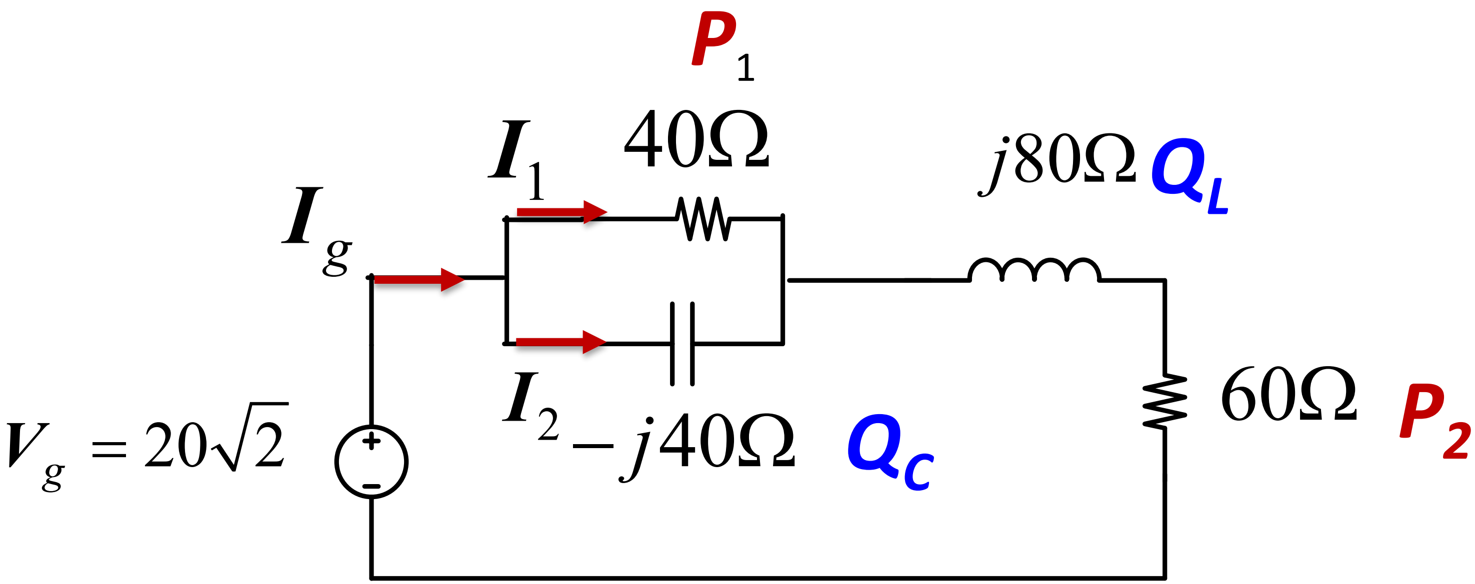
$$P_{dev} = P_1 + P_2$$

$$Q_{dev} = Q_L + Q_C$$



$$\boldsymbol{I}_1 = \boldsymbol{I}_g \frac{(-40j\Omega \,\|\, 40\Omega)}{40\Omega} = (0.0285 - 0.1987j)\,A$$

$$\boldsymbol{I}_2 = \boldsymbol{I}_g - \boldsymbol{I}_1 = (0.1975 + 0.029j)\,A$$



$$P_1 + P_2 = \left|\boldsymbol{I}_1\right|^2 40\Omega + \left|\boldsymbol{I}_g\right|^2 60\Omega$$

$$Q_C + Q_L = -j\left|\boldsymbol{I}_2\right|^2 40\Omega + j\left|\boldsymbol{I}_g\right|^2 80\Omega$$

$$P_{dev} = P_1 + P_2$$

$$Q_{dev} = Q_L + Q_C$$

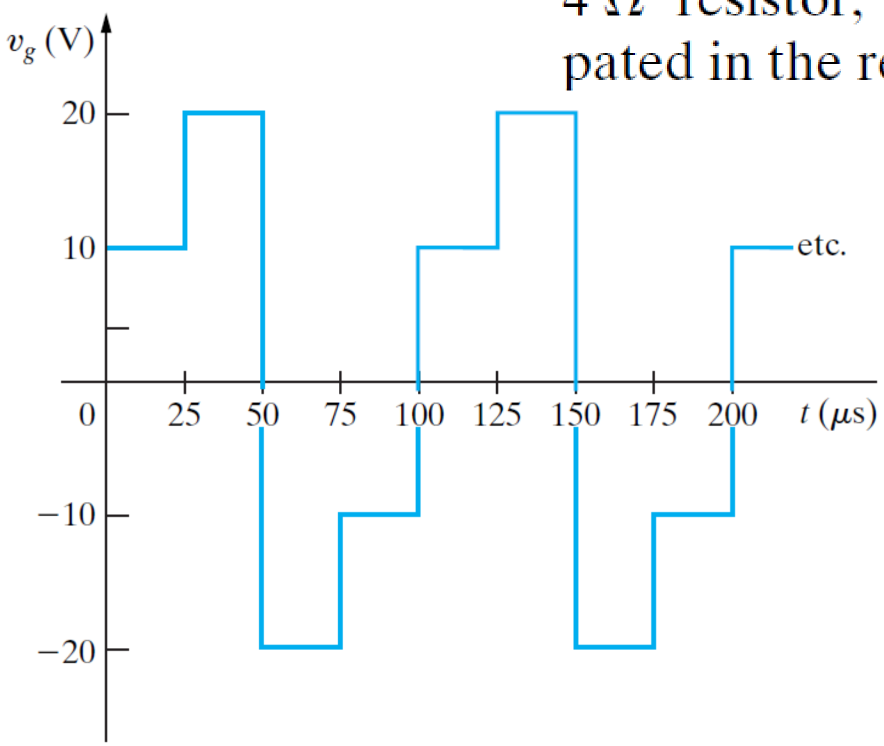
Homework

- P10.15

a) Find the rms value of the periodic voltage shown in Fig. P10.15.

b) If this voltage is applied to the terminals of a 4 Ω resistor, what is the average power dissipated in the resistor?

Figure P10.15



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